WORKED SOLUTION

# STEP II 1990, Question 4

Pure • Line Arrangements and Induction

**STEP II 1990, Question 4**

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/90-s2-q4

This question has a full guided treatment in *Beyond STEP*, Section 9.1, *Lines Cutting the Plane*: https://beyondstep.keithmcnulty.org/ch09/lazy-caterer.html

## The key idea

Look at the picture one line at a time. When a new line is added, it crosses every old line exactly once, at different points, so it is chopped into a predictable number of pieces. Each of those pieces does two jobs: it is a new segment, and it cuts an existing region in two. Counting the effect of one new line gives a recurrence, and the recurrences are easy to sum.

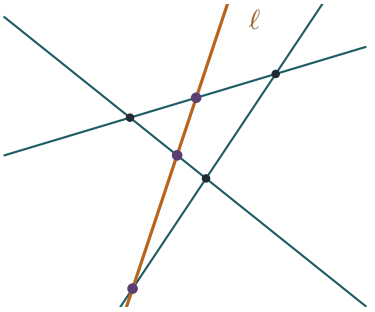
## Solution

### Small cases

Throughout, a "segment" means a piece of one of the lines between consecutive crossing points, including the two infinite pieces (rays) at the ends of each line, and with one line the whole line counts as one segment. Drawing the first few cases gives:

| $n$ | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| segments | 1 | 4 | 9 | 16 |
| regions | 2 | 4 | 7 | 11 |

For instance, three lines form a triangle: each line is cut into a finite side and two rays, giving 9 segments, and the regions are the triangle, three regions touching its sides and three touching its corners, 7 in all. The segment counts suggest $n^2$.




The figure shows three lines (teal) and a fourth line $\ell$ (amber). The three plum points where $\ell$ crosses the old lines cut it into four pieces, and each piece splits one of the seven old regions, giving 11.

#### The number of segments

Fix one of the $n$ lines. No two lines are parallel, so it meets each of the other $n-1$ lines, and since no three lines are concurrent, these $n-1$ crossing points are all different (if two coincided, three lines would meet there). So the line carries $n-1$ distinct points, which cut it into $n$ pieces: $n-2$ finite segments and 2 rays. Doing this for each of the $n$ lines,

$$\text{number of segments} = n \cdot n = n^2.$$

Each segment is counted once, because each lies on exactly one line.

If you prefer induction, let $S(n)$ be the number of segments. Adding an $n$th line to $n-1$ lines in general position, the new line is cut into $n$ pieces, and each of the $n-1$ old lines it crosses has one of its segments split in two, adding one more segment each. So

$$S(n) = S(n-1) + n + (n-1) = S(n-1) + 2n - 1,$$

and with $S(1) = 1$, if $S(n-1) = (n-1)^2$ then $S(n) = (n-1)^2 + 2n - 1 = n^2$.

#### The number of regions

Let $R(n)$ be the number of regions made by $n$ lines in general position, so $R(1) = 2$. Start from $n-1$ such lines and add the $n$th line $\ell$. As above, $\ell$ crosses the old lines at $n-1$ distinct points and is cut into $n$ pieces.

Take one of these pieces. Its interior doesn't meet any old line, so it lies inside a single old region, and it runs right across that region from boundary to boundary (or out to infinity). It therefore splits that region into two.

> **Ask yourself** Could two of the $n$ pieces lie in the same old region, so that fewer than $n$ new regions appear?

No. Each region is an intersection of half-planes (on a fixed side of each old line), so it is convex. A line meets a convex set in a single interval, so $\ell$ passes through each old

region at most once, and different pieces lie in different regions. Also, regions that $\ell$ misses are untouched.

So adding $\ell$ splits exactly $n$ old regions in two and leaves the rest alone:

$$R(n) = R(n-1) + n.$$

Now we use induction. The formula $\frac{1}{2}(n^2+n+2)$ gives 2 when $n=1$, which is correct. If $R(n-1) = \frac{1}{2}\big((n-1)^2+(n-1)+2\big) = \frac{1}{2}(n^2-n+2)$, then

$$R(n) = \tfrac{1}{2}(n^2-n+2) + n = \tfrac{1}{2}(n^2+n+2),$$

so the formula holds for all $n \geq 1$, as required. (Equivalently, $R(n) = 1 + (1+2+\cdots+n) = 1 + \frac{1}{2}n(n+1)$, where the leading 1 is the whole plane with no lines.)

## Checks

- The formula gives $2, 4, 7, 11$ for $n = 1, 2, 3, 4$, matching the drawings.
- Euler's formula gives an independent check. Join the infinite rays at a single "point at infinity". Then there are $V = \binom{n}{2} + 1$ vertices, $E = n^2$ edges and $F$ faces with $V - E + F = 2$, so $F = n^2 - \binom{n}{2} + 1 = \frac{1}{2}(n^2+n+2)$.
- We drew random sets of up to 8 lines with integer coefficients on a computer, counted segments by locating the crossing points, and counted regions as the number of different patterns of "which side of each line" among sample points around every crossing and far away. For example, 8 lines gave 64 segments and 37 regions every time.

## Common slips

- Forgetting the two infinite rays on each line, which gives $n(n-2)$ instead of $n^2$.
- Saying the new line adds $n-1$ regions (one per crossing) rather than $n$ (one per piece).
- Not using the hypotheses: without "no three concurrent" the crossing points on a line need not be distinct, and the counts drop.

## Beyond STEP

This is the simplest case of a *hyperplane arrangement*. The same argument one dimension up shows that $n$ planes in general position cut space into $\frac{1}{6}(n^3+5n+6)$ pieces, because each new plane is itself cut into $R(n-1)$ regions by the old ones. The Euler-formula check is a first glimpse of the Euler characteristic, a central idea in topology. For a lovely chapter on Euler's formula and its uses, see Aigner and Ziegler, *Proofs from THE BOOK*.

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